Lab07. Running application in the cloud.

Cloud Programming (W04IST–SI4527G)

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1 Learning goals

1. How to deploy your application to infrastructure deployed using Terraform.

2 Exercises

In this exercise, your goal is to configure single EC2 instance, then deploy the chat application (backend and frontend) used in the previous labs in this instance.

Prepare Terraform configuration file to build the following configuration.

Do not use VPC module, but configure each resource individually.

1. Create network infrastructure for your EC2 instance:
   - VPC :

     use IP network address 10.X.Y.0/24, where X is index no. of one student modulo 255, and Y is index no. of the other student modulo 255,
   - subnet :

     use network mask with the length of 26 bits,
   - Internet Gateway,
   - route table, including routes,
   - Security Groups (for EC2 instance), including all rules required for your application and remote management.

   Checkpoint: Apply Terraform configuration and verify configuration of your network.
2. Configure EC2 instance in the subnet created in the previous step.

   Checkpoint: Verify if you can remotely access EC2, and access Internet from EC2.
3. Create separate Shell script file, and use it, as user data to configure EC2 instance:
   - install software required to build your backend,

- clone application from your GitHub repository (lab 4, tip: use Personal Access Token [1]),
- build backend,
- configure backend application to run as a Linux service, and start at the boot (tip: see [2]),
- start your backend.

Checkpoint: Verify if your backend is available from the Internet. Confirm that it works after the EC2 reboot.

4. Extend Shell script to:
   - install software required to build your frontend,
   - build frontend (tip: read [3] to learn how to get public IP address in Shell),
   - configure frontend application to run as a Linux service, and start at the boot,
   - start your frontend.

   Checkpoint: Verify if your frontend is available from the Internet.

5. Polish your solution:
   - configure Terraform to display public IP and DNS of the created EC2 instance,
   - confirm, that frontend still works after the EC2 reboot,

2.1 Final report

1. Write a report about your configuration in the README.md file.
   1. Include screenshots for each checkpoint and describe your configuration.
   2. Describe your problems and obstacles encountered in this assignment.

Bibliography

[1] "Creating a fine-grained personal access token." [Online]. Available: https://docs.github.com/en/authentication/keeping-your-account-and-data-secure/managing-your-personal-access-tokens#creating-a-fine-grained-personal-access-token.

[2] "Run a java application as a service on linux." [Online]. Available: https://www.baeldung.com/linux/run-java-application-as-service.

[3] "Use the instance metadata service to access instance metadata." [Online]. Available: https://docs.aws.amazon.com/AWSEC2/latest/UserGuide/configuring-instance-metadata-service.html#instance-metadata-retrieval-examples.